

ARUNAVA GHOSH

24BCG10121

```
import com.mongodb.client.MongoClient;
import com.mongodb.client.MongoClients;
import com.mongodb.client.MongoDatabase;
import com.mongodb.client.MongoCollection;
import com.mongodb.client.model.Filters;
import com.mongodb.client.model.Updates;
import org.bson.Document;

import java.util.Arrays;

public class MongoDBTemplate {

    public static void main(String[] args) {

        String uri = "mongodb://localhost:27017";

        try (MongoClient client = MongoClients.create(uri)) {

            System.out.println("connect");

            MongoDatabase db = client.getDatabase("collegeDB");
            MongoCollection<Document> col = db.getCollection("Students");

            System.out.println("insert");

            col.insertMany(Arrays.asList(
                new Document("sid", 701).append("name", "Ravi").append("course",
"CSE").append("cgpa", 8.2),
                new Document("sid", 702).append("name", "Neha").append("course",
"ECE").append("cgpa", 9.0),
                new Document("sid", 703).append("name", "Aman").append("course",
"AIML").append("cgpa", 8.6)
            ));

            System.out.println("read");

            for (Document d : col.find()) {
                System.out.println(d.toJson());
            }

            System.out.println("find");
            System.out.println(col.find(Filters.eq("sid", 702)).first().toJson());

            System.out.println("update");
            col.updateOne(Filters.eq("sid", 702), Updates.set("name", "Neha
Sharma"));

            System.out.println("replace");
            col.replaceOne(
                Filters.eq("sid", 703),
                new Document("sid", 703).append("name", "Karan").append("course",
"IT").append("cgpa", 9.1)
```

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    );

    System.out.println("delete");
    col.deleteOne(Filters.eq("sid", 701));

    System.out.println("final");

    for (Document d : col.find()) {
        System.out.println(d.toJson());
    }

} catch (Exception e) {
    System.out.println("fail " + e.getMessage());
}

}
}

```

```

~/Developer/uni/dsn2093/face-prep-mongodb-database-administrator/java-dbms λ ls
_deps                               makefile                            MariaDBTemplate.java
build                               MariaDBConnection.java MongoDBTemplate.java
~/Developer/uni/dsn2093/face-prep-mongodb-database-administrator/java-dbms λ make clean
rm -rf build
~/Developer/uni/dsn2093/face-prep-mongodb-database-administrator/java-dbms λ make run-mongo
mkdir -p build
javac -d build -cp ".:_deps/*" *.java
java -cp "build:_deps/*" MongoDBTemplate
Jun 03, 2026 8:38:52 PM com.mongodb.internal.diagnostics.logging.Loggers shouldUseSLF4J
WARNING: SLF4J not found on the classpath. Logging is disabled for the 'org.mongodb.driver' component
connect
insert
read
{"_id": {"$oid": "6a2043844c6a6d7990872a2c"}, "sid": 701, "name": "Ravi", "course": "CSE", "cgpa": 8.2}
{"_id": {"$oid": "6a2043844c6a6d7990872a2d"}, "sid": 702, "name": "Neha", "course": "ECE", "cgpa": 9.0}
{"_id": {"$oid": "6a2043844c6a6d7990872a2e"}, "sid": 703, "name": "Aman", "course": "AIIML", "cgpa": 8.6}
find
{"_id": {"$oid": "6a2043844c6a6d7990872a2d"}, "sid": 702, "name": "Neha", "course": "ECE", "cgpa": 9.0}
update
replace
delete
final
{"_id": {"$oid": "6a2043844c6a6d7990872a2d"}, "sid": 702, "name": "Neha Sharma", "course": "ECE", "cgpa": 9.0}
{"_id": {"$oid": "6a2043844c6a6d7990872a2e"}, "sid": 703, "name": "Karan", "course": "cours
~/Developer/uni/dsn2093/face-prep-mongodb-database-administrator/java-dbms λ |

```

Figure 1: Mongo CRUD

```

MongoDBTemplate.java
5      MongoDBDatabase db = client.getDatabase("collegeDB");
4      MongoCollection<Document> col = db.getCollection("Students");
3
2      System.out.println("insert");
1
0      col.insertMany(Arrays.asList(
1          new Document("sid", 701).append("name", "Ravi").append("course", "CSE").append("cgpa",
2              8.2),
2          new Document("sid", 702).append("name", "Neha").append("course", "ECE").append("cgpa",
3              9.0),
3          new Document("sid", 703).append("name", "Aman").append("course", "AIIML").append("cgpa",
4              8.6)
4      ));
5
6      System.out.println("read");

": 9.0}
update
replace
delete
final
{"_id": {"$oid": "6a2043844c6a6d7990872a2d"}, "sid": 702, "name": "Neha Sharma", "course": "ECE", "cgpa": 9.0}
{"_id": {"$oid": "6a2043844c6a6d7990872a2e"}, "sid": 703, "name": "Karan", "course": "cours
~/Developer/uni/dsn2093/face-prep-mongodb-database-administrator/java-dbms λ

```

Figure 2: Code